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Data-driven adaptive consensus for linear multi-agent systems: A scalable distributed protocol☆

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ABSTRACT

This paper is concerned with the noiseless and noisy data-driven consensus problem of general linear multi-agent systems (MASs) with unknown agent dynamics. First, a data-driven adaptive scheme is designed to enable each edge to tune its weight in an on-line fashion. Subsequently, a distributed noiseless data-driven adaptive consensus (DDAC) protocol is established for the MASs so as to ensure guaranteed consensus. In this protocol, agents communicate with their neighbors through an undirected and connected graph. Importantly, this protocol is proven to be independent of both system model knowledge and be scalable with respect to the size of communication network. Moreover, to address the scenario of a directed communication graph, a modified node-based adaptive scheme, which relies solely on data, is introduced, along with a refined DDAC protocol. The conditions for achieving consensus are derived as semi-definite programs, and the corresponding feasibility is analyzed. Furthermore, the paper considers a noisy data scenario and tackles the consensus problem with a noisy data by employing a refined adaptive scheme and establishing a distributed noisy DDAC protocol. Compared to existing consensus protocols, our DDAC protocol offers high flexibility and scalability by eliminating the need for a system model and global network information. Finally, three examples are provided to verify the effectiveness of the proposed DDAC protocols.

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1. Introduction

In the past few decades, the consensus problem of multi-agent systems (MASs) has attracted much attention from a variety of engineering fields owing to its clear application potentials in various engineering areas such as sensor networks (Olfati-Saber, 2007), spacecraft formation flying (Han, Guo, Xie, & Lin, 2019), autonomous underwater vehicles (Linsenmayer, Dimarogonas, & Allgöwer, 2017), and mobile robots (Lu, Yu, Chen, & Yu, 2019). To address the consensus problem, one key task is to construct an effective distributed consensus protocol to properly adjust the behavior of agents (Ding, Han, Ge, & Zhang, 2018; Han et al., 2019; Qin, Ma, Yu, & Kang, 2018; Yang, Meng, Dimarogonas, & Johansson, 2014). Until now, a substantial body of literature has been devoted to the design of consensus protocols for MASs with various dynamics (such as single-integrator dynamics (Xu, Hu, Ho, & Feng, 2020), double-integrator dynamics (Ren, 2008), linear and nonlinear dynamics (Li, Wen, Duan, & Ren, 2015; Xu, He, Ho, & Kurths, 2022)) and with different communication structures (such as undirected or directed graphs (Ren, 2008; Yu, Chen, Cao, & Kurths, 2010)).

It should be noted that the majority of existing consensus protocols adopt a *model-based* approach relying on a model of agent dynamics for protocol design (Han et al., 2019; He, Qian, Han, & Chen, 2020; Li et al., 2015; Olfati-Saber, 2007; Xu et al., 2022). However, in practical MASs, obtaining or establishing an accurate mathematical model can be challenging or even impossible. Therefore, achieving consensus behavior while controlling MASs with *unknown* dynamics becomes a significant research problem. To address this challenge, several model-free methods have been proposed to utilize measured data to replace or recover model information. Some well-established model-free methods include, but are not limited to, system identification (Hou & Wang, 2013; Verhaegen & Verdult, 2007), adaptive dynamic programming (Bu, Hou, & Zhang, 2017; Zhang, Yue, Dou, Zhao, & Xie, 2019), and neural network learning (Gao, Wang, & Yu, 2021; Yang, Zhang, & Wang, 2022; Zhang, Jiang, Luo, & Xiao, 2016). These methods aim to first identify a system model and then design a robust control law by adopting classical model-based approaches.

In addition to the aforementioned model-free methods, there has been recent research interest in a direct data-driven control

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technique (Bianchi, Grammatico, & Cortés, 2025; Campi, Lecchini, & Savaresi, 2002; De Persis & Tesi, 2019; Karimi, van Heusden, & Bonvin, 2007; Van Waarde, Eising, Trentelman, & Camlibel, 2020), which has gained attention in the field. This approach bypasses the system identification process and aims to directly control system performance using data. Compared to traditional model-free methods that combine system identification and model-based control, the direct data-driven control technique offers the following advantages: 1) it reduces the computational complexity associated with system identification, and 2) it requires less system information than fully identifying the system model (Van Waarde et al., 2020). Due to these distinct advantages, the direct data-driven control technique has been employed to address the stabilization problem in control systems (Bianchi et al., 2025; De Persis, Postoyan, & Tesi, 2024; De Persis & Tesi, 2019; Van Waarde et al., 2020; Wang et al., 2023). However, despite its notable benefits, the direct data-driven control approach has received little attention when it comes to the consensus issue of MAS, and such a research gap serves as the motivation for our current investigation.

In certain applications such as mobile robots, agents may have limited sensing capabilities, and consensus protocols should rely solely on local information from each agent and its neighbors. However, designing a protocol inevitably requires some global information such as the smallest nonzero eigenvalue of the Laplacian matrix associated with the communication graph. Obtaining or computing such global information can be extremely challenging, particularly in large-scale or open networks. Moreover, the involvement of global information hinders the scalability and flexibility of control protocols, diminishing the advantages of distributed control. To address these limitations, an emerging trend in the field of MAS consensus is the development of adaptive strategies that allow for the tuning of coupling gains without the need for knowledge or manipulation of the Laplacian matrix. Noteworthy works in this area include (Li, Ren, Liu, & Xie, 2013; Li, Sun, Tang, & Karimi, 2020; Li et al., 2015; Wang, Ren, Yu, & Zhang, 2021; Xu et al., 2022). However, it is important to mention that the adaptive schemes proposed in these works require explicit knowledge of the system models, making them unsuitable for MASs with unknown dynamics.

Motivated by the previous discussion, this paper aims to co-design the consensus protocol and adaptive law directly using available data, enabling MASs with unknown agent dynamics to achieve the desired asymptotic consensus in a scalable distributed manner. The main contributions of this paper can be summarized as follows.

(1) A novel direct data-driven approach with noiseless/noisy data is proposed to design a scalable distributed consensus protocol for MASs with unknown general linear dynamics, marking the first of its kind.
(2) Both noiseless and noisy data-driven adaptive consensus (DDAC) protocols are established, where the control gain and adaptive law are directly designed with data, thereby eliminating the need for explicit identification of system matrices.
(3) Two distinct adaptive laws are introduced, based on edge and node information, respectively, tailored for undirected and directed communication graphs. These adaptive laws allow for online tuning of the coupling gains without requiring knowledge of the Laplacian matrix's eigenvalues.
(4) Sufficient conditions are derived for achieving consensus in the form of semi-definite programs (SDP) or linear matrix inequalities (LMIs) so as to ensure feasibility in implementation.

In summary, the purpose of this paper is to pave the way for achieving consensus in MASs with unknown agent dynamics through a direct data-driven approach that offers practical and scalable solutions.

Notation. The notation used here is fairly standard except where otherwise stated. $\mathbb{R}^{n\times m}$ and $\mathbb{R}^n$ denote, respectively, the set of $n\times m$ real matrices and n-dimensional Euclidean space. $\boldsymbol{I}_{n\times m}$ and $\mathbb{O}_{n\times m}$ denote the sets of $n\times m$ matrices with all elements being 1 and 0, respectively. For matrices A and B, their Kronecker product is denoted as $A\otimes B$. diag$\{\cdots\}$ stands for a block-diagonal matrix with the main-diagonal blocks being square matrices and all off-diagonal blocks being zero matrices. For vectors $x_1,\ldots,x_n$, col$(x_1,\ldots,x_n)\triangleq[x_1^\top,\ldots,x_n^\top]^\top$. For a real matrix A, $A\succ(\succeq)0$ means symmetric and positive (semi-)definite, and $A\prec(\preceq)0$ means symmetric and negative (semi-)definite.

2. Preliminaries

In this section, we provide a review of fundamental results from algebraic graph theory and present the formulation of the consensus problem for MASs, which will serve as the basis for the subsequent analysis in this paper.

2.1. Graph theory

Let a directed graph be $\mathcal{G}=(\mathcal{V},\mathcal{E})$ with a node set $\mathcal{V}=\{1,2,\ldots,N\}$ and an edge set $\mathcal{E}\subseteq\mathcal{V}\times\mathcal{V}$. If there exists a directed edge from nodes i to j, then the node j is a *neighbor* of node i and let $(i,j)\in\mathcal{E}$. For an *undirected* graph $\mathcal{G}$, $(i,j)\in\mathcal{E}\Leftrightarrow(j,i)\in\mathcal{E}$. Here, $(i,i)\notin\mathcal{E}$ for $\forall i\in\mathcal{V}$. Let the set $\mathcal{N}_i=\{j:(i,j)\in\mathcal{E}\}$ include all neighbors of node i, and d_i be the total number of neighbors. A directed path from nodes i to j is a sequence of edges $(i,j_1),(j_1,j_2),\cdots,(j_m,j)$ in graph $\mathcal{G}$ with distinct nodes j_k $(k=1,2,\ldots,m)$. The undirected (directed) graph $\mathcal{G}$ is (strongly) connected if and only if there exists a directed path between arbitrary two agents.

Let $\mathcal{H}=[h_{ij}]\in\mathbb{R}^{N\times N}$ be the adjacency matrix with nonnegative elements

$$\forall i\in\mathcal{V},\quad h_{ij}=\begin{cases}1 & j\in\mathcal{N}_i,\\ 0 & \text{otherwise},\end{cases}\tag{1}$$

and let $\mathcal{L}=[l_{ij}]\in\mathbb{R}^{N\times N}$ be the Laplace matrix with

$$\forall i\in\mathcal{V},\quad l_{ij}=\begin{cases}-h_{ij} & i\neq j,\\ \sum_{k\in\mathcal{N}_i}h_{ik} & i=j.\end{cases}\tag{2}$$

Lemma 1 (Horn & Johnson, 2012). *If graph $\mathcal{G}$ is undirected and connected, then the corresponding Laplacian matrix $\mathcal{L}$ has a simple eigenvalue 0 and the other eigenvalues are all positive.*

Lemma 2 (Yu et al., 2010). *If graph $\mathcal{G}$ is directed and strongly connected, then there exists an eigenvector $\zeta\triangleq\text{col}\{\zeta_i\}$ with $\zeta_i>0$ such that $\zeta^\top\mathcal{L}=0$ and $\sum_{i=1}^N\zeta_i=1$. In addition,*

$$\min_{\zeta^\top x=0}x^\top(\Theta\mathcal{L}+\mathcal{L}^\top\Theta)x\geq\lambda_*x^\top x$$

where $\Theta=\text{diag}\{\zeta_i\}$ and λ_* is the second smallest eigenvalue of the matrix $\Theta\mathcal{L}+\mathcal{L}^\top\Theta$.

2.2. Description of linear MAS and model-based consensus protocol

Consider an MAS with N agents whose dynamics can be described by the following model:

$$\dot{x}_i(t)=Ax_i(t)+Bu_i(t),\quad i\in\mathcal{V}\tag{3}$$

where $x_i(t)\in\mathbb{R}^n$ and $u_i(t)\in\mathbb{R}^m$ denote the state and the control input of agent i at time instant t, respectively. $A\in\mathbb{R}^{n\times n}$ and $B\in\mathbb{R}^{n\times m}$ are constant matrices, and the pair (A,B) is stabilizable. In this paper, the matrices A and B are assumed to be unknown.

Example 1. When A and B are known, there have been abundant results to deal with the asymptotic consensus problem. For example, in Olfati-Saber and Murray (2004), a classical model-based control protocol with relative states is given as

$$u_i(t) = K \sum_{j=1}^{N} h_{ij} \left(x_i(t) - x_j(t)\right) \tag{4}$$

where the control gain is design by $K = -B^\top P$ with P being the solution to the following algebraic Riccati equation (ARE)

$$A^\top P + PA - \mu_1 PBB^\top P + \mu_2 I = 0 \tag{5}$$

with $\mu_i > 0$ $(i = 1, 2)$. For example, in Ni and Cheng (2010), $\mu_1 = 2\mu_2 = 2\rho_*$ and in Hu, Liu, and Feng (2016), $0 < \mu_1 \leq 2\rho_*$ and $\mu_2 > 2\rho^*$ where ρ_* and ρ^* are the smallest and largest nonzero eigenvalue of Laplacian matrix. Obviously, the gain K is explicitly dependent not only on the system matrices A and B, but also on the network topology.

This paper focuses on the scenario where the matrices A and B are unknown. It is important to note that the existing methods, which rely on known A and B, are not applicable to this case. Therefore, new approaches need to be developed to tackle the consensus problem in the presence of unknown matrices.

To address the asymptotic consensus problem for the general linear MAS described by (3) with unknown dynamics, we aim to develop a consensus protocol u_i that achieves $\lim_{t\to\infty} x_i(t) - x_j(t) = 0$ $(\forall i, j \in \mathcal{V})$ without relying on the knowledge of system matrices A and B. The objective is to design an effective consensus protocol that can facilitate the convergence of the agent states to a common value despite the unknown dynamics.

In the case of unknown matrices A and B, one approach is to first identify the system model and then design a consensus protocol using classical model-based approaches. However, in this paper, we propose a new perspective to address the consensus problem of unknown MASs. We bypass the process of system identification and, instead, design the consensus protocol directly from available data. This data-driven approach allows us to design a consensus protocol without explicitly identifying the system model, offering a novel and alternative method to achieve consensus in MASs with unknown dynamics.

The main aim of this paper is to co-design a consensus protocol and an adaptive scheme directly using available data, which enable MASs with unknown dynamics to achieve asymptotic consensus. The specific objectives of this research can be summarized as follows.

(1) *Design a novel data-driven adaptive scheme that utilizes a coupling gain represented by data matrices*. This scheme allows for on-line tuning of the weights of each edge without requiring knowledge of the Laplacian matrix or the system model.
(2) *Develop distributed edge/node-based DDAC protocols directly from available noiseless or noisy data*. These protocols are designed to be implemented in a scalable distributed manner, via bypassing the system identification process, with hope to enable MASs with unknown dynamics to achieve asymptotic consensus through undirected/directed communication graphs.

3. Learning DDAC protocol with noise-free data

In this section, we introduce a direct data-driven approach for designing the control gain and coupling matrix using *noise-free* data in order to establish DDAC protocols, which ensure the asymptotic consensus of the MAS (3) under both undirected and directed communication graphs. Additionally, we derive sufficient conditions that guarantee the asymptotic consensus. By employing this direct data-driven method, we eliminate the need for system identification and provide a framework for designing robust and scalable consensus protocols in MASs with unknown dynamics.

3.1. Dataset training

To design a consensus protocol from data, we adopt the data-based representation developed in De Persis and Tesi (2019), Willems, Rapisarda, Markovsky, and De Moor (2005) for the system (3). Suppose that sampled measurements $\mathcal{D}_i \triangleq \{u_i(s\cdot T_p), x_i(s\cdot T_p), \dot{x}_i(s\cdot T_p)\}_{s=0}^{S-1}$ of agent $i \in \mathcal{V}$ in (3) can be collected off-line at discrete time-instants $\{s\cdot T_p\}_{s=0}^{S-1}$, which consist of input, state and state derivative data from any agent i using an open-loop experiment, where T_p is a sampling time used for collecting the data and $S > 0$ is the number of samples.

According to (3), each agent is assumed to have the same system matrices A and B. Without loss of generality, we can choose one leading agent (labeled as d) that collects these measurements in $\mathcal{D}_d$ off-line to form the following data matrices

$$\begin{aligned}
\mathcal{U}^d &\triangleq [u_d(0), u_d(T_p), \ldots, u_d((S-1)T_p)] \in \mathbb{R}^{m\times S}\\
\mathcal{X}_0^d &\triangleq [x_d(0), x_d(T_p), \ldots, x_d((S-1)T_p)] \in \mathbb{R}^{n\times S}\\
\mathcal{X}_1^d &\triangleq [\dot{x}_d(0), \dot{x}_d(T_p), \ldots, \dot{x}_d((S-1)T_p)] \in \mathbb{R}^{n\times S}
\end{aligned} \tag{6}$$

with

$$\dot{x}_d(s\cdot T_p) = Ax_d(s\cdot T_p) + Bu_d(s\cdot T_p).$$

Obviously, the following is true:

$$\mathcal{X}_1^d = A\mathcal{X}_0^d + B\mathcal{U}^d. \tag{7}$$

Assumption 1. The matrices $\mathcal{U}^d$, $\mathcal{X}_0^d$, and $\mathcal{X}_1^d$ are known.

Assumption 2. The matrix $\begin{bmatrix}\mathcal{U}^d\\ \mathcal{X}_0^d\end{bmatrix}$ has full row rank.

Assumption 2, which can be easily checked for a given set $\mathcal{D}_d$, is actually on the richness of the data in $\mathcal{D}_d$.

Lemma 3 (*De Persis et al., 2024*)**.** *Under Assumption 2, for any matrix $K \in \mathbb{R}^{m\times n}$, there exists a matrix $G \in \mathbb{R}^{S\times n}$ such that*

$$\begin{bmatrix}K\\ I_n\end{bmatrix} = \begin{bmatrix}\mathcal{U}^d\\ \mathcal{X}_0^d\end{bmatrix} G. \tag{8}$$

In this case, $A + BK = \mathcal{X}_1^d G$.

Under Assumption 2, it is easy to obtain the following lemma.

Lemma 4. *Under Assumption 2, for any matrix $K \in \mathbb{R}^{m\times n}$, there exists a matrix $L \in \mathbb{R}^{S\times n}$ such that*

$$\begin{bmatrix}K\\ \mathbb{O}_n\end{bmatrix} = \begin{bmatrix}\mathcal{U}^d\\ \mathcal{X}_0^d\end{bmatrix} L. \tag{9}$$

In this case, $BK = \mathcal{X}_1^d L$.

To facilitate the analysis, we introduce two matrices:

$$M_d \triangleq \begin{bmatrix}\mathcal{U}^{d\top} & \mathcal{X}_0^{d\top}\end{bmatrix}, \quad \mathbb{I}_d \triangleq \begin{bmatrix}\boldsymbol{I}_{m\times m}\\ \mathbb{O}_{n\times m}\end{bmatrix}. \tag{10}$$

Under Assumption 2, the matrix $M_d^\top M_d$ is invertible. Denote $M_v \triangleq (M_d^\top M_d)^{-1}$.

3.2. DDAC protocol design under undirected graph

In this subsection, we discuss the design of noiseless DDAC protocol under the undirected and connected communication graph $\mathcal{G}$.

Assumption 3. The communication graph $\mathcal{G}$ is undirected and connected.

A distributed edge-based DDAC protocol is designed as follows:

$$u_i(t) = K_d \sum_{j=1}^{N} s_{ij}(t) h_{ij} \left(x_i(t) - x_j(t)\right) \tag{11a}$$

$$\dot{s}_{ij}(t) = p_{ij} h_{ij} (x_i(t) - x_j(t))^\top \Gamma_d (x_i(t) - x_j(t)) \tag{11b}$$

and the control gain K_d and the coupling matrix Γ_d are designed in the form of

$$K_d = \mathcal{U}^d Y (\mathcal{X}_0^d Y)^{-1}, \tag{12a}$$

$$L_d = M_d M_v \mathbb{I}_d K_d, \tag{12b}$$

$$\Gamma_d = -(\mathcal{X}_0^d Y)^{-1} \mathcal{X}_1^d L_d - (\mathcal{X}_1^d L_d)^\top (\mathcal{X}_0^d Y)^{-1} \tag{12c}$$

where h_{ij} is the (i,j)-th entry of $\mathcal{H}$ with respect to $\mathcal{G}$, and $s_{ij}(t)$ is time-varying coupling weights. $s_{ij}(0) = s_{ji}(0) > 0$ if $(i,j) \in \mathcal{E}$, and $s_{ij}(0) = s_{ji}(0) = 0$ otherwise. The weights $p_{ij} = p_{ji} > 0$ can be arbitrarily chosen. The matrix $Y \in \mathbb{R}^{S \times n}$ will be determined later.

Note that (12) provides a purely data-driven design of the control gain K_d and coupling gain Γ_d using only the data $\mathcal{U}^d$, $\mathcal{X}_0^d$, and $\mathcal{X}_1^d$. In this paper, the agent d undertakes the responsibility of computing the control gains K_d and Γ_d off-line, subsequently transmitting them to its neighboring agents. Employing finite-step transmission, all agents can successfully receive K_d and Γ_d due to the connected/strongly-connected network. The above process will be carried out in the initial stage. Notably, K_d and Γ_d remain fixed even as the network topology evolves.

Remark 1. In most existing consensus protocols, the control gains are determined based on global network information such as the eigenvalues of the Laplacian matrix. Unfortunately, this approach has limitations in terms of scalability when modifying network structures. To address this limitation and design a scalable distributed consensus protocol independent of global network information, adaptive schemes have been introduced to dynamically adjust control gains. It is worth noting that all these adaptive schemes rely on explicit system models. For example, in Li et al. (2013, 2020, 2015), Wang et al. (2021), Xu et al. (2022), the coupling matrix Γ is defined as $\Gamma \triangleq P^{-1} B B^\top P^{-1}$ where B represents the knowledge of the system model and P is determined by the system matrices A and B. However, accurately identifying the system matrices A and B can be challenging in practical applications. To address this issue, we propose a new data-driven adaptive scheme (11b) for each edge in the consensus protocol. In this scheme, the matrix Γ_d in (12c) is designed using data matrices instead of relying on explicit model knowledge. This data-driven approach allows for adaptive tuning of the coupling matrix without the need for accurate identification of the system matrices A and B.

Remark 2. The control gain K_d and coupling gain Γ_d can be computed offline using the data matrices $\mathcal{U}^d$, $\mathcal{X}_0^d$, and $\mathcal{X}_1^d$. Once computed, these gains can be shared with all agents through the communication network. Importantly, there is no need to recalculate the gains K_d and Γ_d during the protocol implementation. Furthermore, the DDAC protocol (11) exhibits high scalability and flexibility. When the scale or structure of the communication topology changes, there is no requirement to re-compute the gains K_d and Γ_d. This is because the adaptive weights s_{ij} are utilized in this paper, allowing for online adjustment as needed. As a result, the DDAC protocol (11) can be considered highly scalable and flexible, accommodating changes in network scale or structure without the need for re-computation of the control and coupling gains.

Define two track error functions $\xi_i(t) = x_i(t) - \frac{1}{N}\sum_{j=1}^{N} x_j(t)$ $(\forall i \in \mathcal{V})$ and $\psi_{ij}(t) = x_i(t) - x_j(t)$ $(\forall i,j \in \mathcal{V})$. According to (3) and (11), one has

$$\dot{\xi}_i(t) = A\xi_i(t) + BK_d \sum_{j=1}^{N} s_{ij}(t) h_{ij} \psi_{ij}(t). \tag{13}$$

Theorem 5. *Consider a general linear MAS (3). Suppose that Assumptions 2–3 hold. Let the decision variable $Y \in \mathbb{R}^{S\times n}$ satisfy the following SDP (14) with the form of*

$$\mathcal{X}_1^d Y + (\mathcal{X}_1^d Y)^\top \prec 0 \tag{14a}$$

$$\mathcal{X}_0^d Y \succ 0 \tag{14b}$$

$$\mathcal{M} Y + (\mathcal{M} Y)^\top \preceq 0 \tag{14c}$$

where $\mathcal{M} = \mathcal{X}_1^d M_d M_v \mathbb{I}_d \mathcal{U}^d$. Then, for any initial condition, the DDAC protocol (11) guarantees that the MAS (3) with general linear dynamics achieves asymptotic consensus.

Proof. Let $P_d \triangleq (\mathcal{X}_0^d Y)^{-1} \succ 0$ by (14b). It follows from Lemma 1 that Assumption 3 guarantees the second smallest eigenvalue $\lambda_2 > 0$ of $\mathcal{L}$. Consider a Lyapunov function candidate as

$$V = \sum_{i=1}^{N} \xi_i^\top P_d \xi_i + \sum_{i=1}^{N}\sum_{j=1}^{N} h_{ij} \frac{(s_{ij} - s_0)^2}{4p_{ij}} \tag{15}$$

with s_{ij} and p_{ij} given in (11a)–(11b) and a positive number s_0 satisfying $s_0\lambda_2 > 1$. It follows from (3) and (6) that

$$\mathcal{X}_1^d = A\mathcal{X}_0^d + B\mathcal{U}^d = \begin{bmatrix} B & A \end{bmatrix} M_d^\top$$

with M_d given in (10). From (10) and (12b), one has

$$\mathcal{X}_1^d L_d = \begin{bmatrix} B & A \end{bmatrix} M_d^\top M_d M_v \mathbb{I}_d K_d = BK_d. \tag{16}$$

It follows from $\mathcal{M}Y + (\mathcal{M}Y)^\top \preceq 0$ and $\mathcal{X}_0^d Y \succ 0$ that

$$(\mathcal{X}_0^d Y)^{-1} \mathcal{M} Y (\mathcal{X}_0^d Y)^{-1} + (\mathcal{X}_0^d Y)^{-1} (\mathcal{M}Y)^\top (\mathcal{X}_0^d Y)^{-1} \preceq 0.$$

In addition, according to the definition of $\mathcal{M}$ and L_d in (12b), one has

$$\mathcal{M} Y (\mathcal{X}_0^d Y)^{-1} = \mathcal{X}_1^d M_d M_v \mathbb{I}_d \mathcal{U}^d Y (\mathcal{X}_0^d Y)^{-1} = \mathcal{X}_1^d L_d,$$

which implies from (12c) that $\Gamma_d \succeq 0$.

The derivative of V along the trajectory of the system (3) is obtained as

$$\begin{aligned} \dot{V} = & \sum_{i=1}^{N} [\xi_i^\top (P_d A + A^\top P_d) \xi_i \\ & + \xi_i^\top (P_d B K_d + K_d^\top B^\top P_d) \sum_{j=1}^{N} s_{ij} h_{ij} \psi_{ij}] \\ & + \sum_{i=1}^{N}\sum_{j=1}^{N} h_{ij} \frac{(s_{ij} - s_0)}{2} \psi_{ij}^\top \Gamma_d \psi_{ij}. \end{aligned} \tag{17}$$

For the undirected graph $\mathcal{G}$, $h_{ij} = h_{ji}$. By (4), one has $s_{ij} = s_{ji}$ due to $s_{ij}(0) = s_{ji}(0)$, and therefore

$$\begin{aligned} & \sum_{i=1}^{N} \xi_i^\top (P_d B K_d + K_d^\top B^\top P_d) \sum_{j=1}^{N} s_{ij} h_{ij} \left(\xi_i - \xi_j\right) \\ = & \frac{1}{2} \sum_{i=1}^{N}\sum_{j=1}^{N} s_{ij} h_{ij} \xi_{ij}^\top (P_d B K_d + K_d^\top B^\top P_d) \xi_{ij}. \end{aligned} \tag{18}$$

It follows from (12c), (14b) and (16) that

$$P_d B K_d + K_d^\top B^\top P_d = (\mathcal{X}_0^d Y)^{-1} B K_d + K_d^\top B^\top (\mathcal{X}_0^d Y)^{-1}, \tag{19}$$

that is,

$$P_dBK_d + K_d^\top B^\top P_d = -\Gamma_d. \tag{20}$$

By substituting (18)–(19) into (17), we obtain

$$\begin{aligned}\dot V =& \sum_{i=1}^N \xi_i^\top (A^\top P_d + P_d A)\xi_i \\ &+ s_0 \xi^\top [\mathcal{L} \otimes (P_dBK_d + K_d^\top B^\top P_d)]\xi. \end{aligned}\tag{21}$$

Since $\mathcal{X}_0^d Y$ is invertible and $\Gamma_d \succeq 0$, we have

$$(\mathcal{X}_0^d Y)^{-1}\mathcal{X}_1^d L + (\mathcal{X}_1^d L)^\top (\mathcal{X}_0^d Y)^{-1} \prec 0,$$

which is equivalent to $P_dBK_d + K_d^\top B^\top P_d \prec 0$. Then, (21) can rewritten as

$$\begin{aligned}\dot V \le& \sum_{i=1}^N \xi_i^\top [A^\top P_d + P_d A + s_0\lambda_2 (P_dBK_d + K_d^\top B^\top P_d)]\xi_i \\ \le& \sum_{i=1}^N \xi_i^\top [(A+BK_d)^\top P_d + P_d(A+BK_d)]\xi_i.\end{aligned}$$

where the second inequality holds due to $s_0\lambda_2 > 1$.

Letting $G = YP_d$, we have $K_d = \mathcal{U}^d G$ and $\mathcal{X}_0^d G = I$, which satisfy (8) and thus $A + BK_d = \mathcal{X}_1^d G$ according to Lemma 3.

It follows from (14a) that $\mathcal{X}_1^d G P_d^{-1} + (\mathcal{X}_1^d G P_d^{-1})^\top \prec 0$ and, equivalently, $P_d\mathcal{X}_1^d G + (\mathcal{X}_1^d G)^\top P_d \prec 0$, which means that $(A + BK_d)^\top P_d + P_d(A+BK_d) \prec 0$. Hence, there exists a positive number μ_1 such that $\dot V \le -\mu_1 \sum_{i=1}^N \xi_i^\top \xi_i$, which implies $\dot V \le 0$ and thus V is monotonously non-decreasing. Note that $\dot V = 0$ if and only if $\xi_i = 0$ for $i = 1, \dots, N$, which implies $\lim_{t\to+\infty} \xi_i(t) = 0$, and the asymptotic consensus can be achieved eventually.

Remark 3. Note that the SDP (14) is feasible. Since (A, B) is stabilizable, there exists a matrix $P_d \succ 0$ such that

$$A^\top P_d + P_d A - 2P_dBB^\top P_d + I = 0. \tag{22}$$

In this case, there exist matrices K_d and $P_d \succ 0$ such that

$$(A+BK_d)^\top P_d + P_d(A+BK_d) \prec 0 \tag{23a}$$

$$P_dBK_d + (BK_d)^\top P_d \preceq 0, \tag{23b}$$

because (23a)–(23b) hold by letting $K_d = -B^\top P_d$. By Lemma 3, for any matrix K_d, there exists a matrix G such that $A + BK_d = \mathcal{X}_1^d G$. Let $Y = GP_d^{-1}$. Obviously, (14a) holds. In addition, it follows from (14b) that $K_d = \mathcal{U}^d G = \mathcal{U}^d Y P_d$ and $I = \mathcal{X}_0^d G = \mathcal{X}_0^d Y P_d$, which leads to $\mathcal{X}_0^d Y = P_d^{-1} \succ 0$ and thus (14b) holds. Moreover, by Lemma 4, there exists a matrix L_d such that $BK_d = \mathcal{X}_1 L_d$. Noticing that $\mathcal{M}Y(\mathcal{X}_0^d Y)^{-1} = \mathcal{X}_1 L_d$, (23b) is equivalent to

$$P_d\mathcal{M}Y(\mathcal{X}_0^d Y)^{-1} + (\mathcal{M}Y(\mathcal{X}_0^d Y)^{-1})^\top P_d \preceq 0, \tag{24}$$

that is, (14c) holds.

3.3. DDAC protocol design under directed graph

In this subsection, we further address the case of a directed graph, where the communication between agents is *not necessarily* bidirectional. To handle this scenario, we propose a modified DDAC protocol that incorporates a node-based adaptive scheme.

Assumption 4. The communication graph $\mathcal{G}$ is directed and strongly-connected.

Let $\psi_i \triangleq \sum_{j=1}^N h_{ij}(x_i - x_j)$ $(\forall i \in \mathcal{V})$ and $\psi \triangleq \mathrm{col}(\psi_i)$. For directed graph $\mathcal{G}$, a modified DDAC protocol is constructed as follows:

$$u_i = K_d s_i r_i(\phi_i) \sum_{j=1}^N h_{ij}\psi_i \tag{25a}$$

$$\dot s_i = p_i \psi_i^\top \Gamma_d \psi_i \tag{25b}$$

where $p_i \in (0, \frac{1}{2})$, $\phi_i \triangleq \psi_i^\top P_d \psi_i$, and $r_i(\phi_i) \triangleq (1+\phi_i)^{\frac{16}{15}}$. The gains K_d and Γ_d are designed in (12) with available data $\mathcal{X}_0^d$, $\mathcal{X}_1^d$ and $\mathcal{U}_0^d$, where P_d and Y will be given subsequently.

Next, (3) and (25) can be rewritten as

$$\dot\psi = [I_N \otimes A + (\mathcal{L}\mathcal{S}\mathcal{R}) \otimes (BK_d)]\psi \tag{26}$$

with $\mathcal{S} = \mathrm{diag}\{s_i\}_{i\in\mathcal{V}}$ and $\mathcal{R} = \mathrm{diag}\{r_i\}_{i\in\mathcal{V}}$.

Theorem 6. *Consider a general linear MAS (3). Suppose that Assumptions 2 and 4 hold. Let $P_d = (\mathcal{X}_0^d Y)^{-1}$ and the decision variable $Y \in \mathbb{R}^{S\times n}$ satisfy the following SDP (14)' with (14a)–(14b) and (14c') given by*

$$\mathcal{M}Y = (\mathcal{M}Y)^\top \preceq 0. \tag{14c'}$$

Then, for any initial condition, the DDAC protocol (25) guarantees that the MAS (3) with general linear dynamics achieves asymptotic consensus.

Proof. Let $P_d = (\mathcal{X}_0^d Y)^{-1} \succ 0$ by (14b). It follows from Lemma 2 that there exists an eigenvector $\zeta \triangleq \mathrm{col}\{\zeta_i\}$ with $\zeta_i > 0$ such that $\zeta^\top \mathcal{L} = 0$ and $\sum_{i=1}^N \zeta_i = 1$. Then, consider the following Lyapunov function candidate

$$V = \underbrace{\sum_{i=1}^N s_i\zeta_i \int_0^{\phi_i} r_i(u)du}_{\triangleq V_1} + \underbrace{\sum_{i=1}^N \frac{\lambda_*(s_i - s_0)^2}{16p_i}}_{\triangleq V_2}. \tag{27}$$

where λ_* is the second smallest eigenvalue of $\Theta\mathcal{L} + \mathcal{L}^\top\Theta$ with $\Theta = \mathrm{diag}\{\zeta_i\}$, and $s_0 > \frac{\zeta_{\max}^4}{\lambda_*^4} + \frac{2\zeta_{\max}}{\lambda_*}$ with $\zeta_{\max} = \{\zeta_i : i \in \mathcal{V}\}$.

It follows from the definition of $\mathcal{M}$ in (14)' and K_d, L_d in (12a)–(12b) that $P_dBK_d = (BK_d)^\top P_d \preceq 0$. Then, there exists a matrix U_d such that $U_d^\top U_d = -P_dBK_d$.

By taking the derivative of V_1 along the trajectory of the system (3), one has

$$\begin{aligned}\dot V_1 =& \sum_{i=1}^N \zeta_i s_i r_i(\phi_i)[\psi_i^\top P\dot\psi_i + \dot\psi_i^\top P\psi_i] \\ &+ \sum_{i=1}^N \zeta_i \dot s_i \int_0^{\phi_i} r_i(u)du \\ \le& \psi^\top [(\mathcal{S}\mathcal{R}\Theta) \otimes (PA + A^\top P) \\ &+ (\mathcal{R}\mathcal{S}(\Theta\mathcal{L} + \mathcal{L}^\top\Theta)\mathcal{S}\mathcal{R}) \otimes (P_dBK_d)]\psi \\ &+ \sum_{i=1}^N \zeta_i \dot s_i r_i(\phi_i)\phi_i \end{aligned}\tag{28}$$

where the inequality holds since r_i are monotonically increasing.

Note that $\xi^\top[(S\mathcal{R}) \otimes U_d]\psi = \xi^\top[(S\mathcal{R}) \otimes U_d](\mathcal{L} \otimes I_n)x = \xi^\top(\mathcal{L} \otimes I_n)[(S\mathcal{R}) \otimes U_d]x = 0$, and thus

$$\begin{aligned}\dot V_1 \le& \psi^\top [(\mathcal{S}\mathcal{R}\Theta) \otimes (P_dA + A^\top P_d) \\ &- (\lambda_*\mathcal{R}^2\mathcal{S}^2) \otimes (U_d^\top U_d)]\psi \\ &+ \sum_{i=1}^N \dot s_i(t) \frac{1}{4\lambda_*^3}\zeta_i^4 + \sum_{i=1}^N \dot s_i \lambda_* \frac{3}{4} r_i^{\frac{4}{3}} \phi_i(t)^{\frac{4}{3}} \\ \le& \psi^\top [(\mathcal{S}\mathcal{R}\Theta) \otimes (P_dA + A^\top P_d)]\psi \\ &- \sum_{i=1}^N \lambda_* s_i^2 r_i^2 \psi_i^\top U_d^\top U_d \psi_i + \sum_{i=1}^N \dot s_i(t) \frac{1}{4\lambda_*^3}\zeta_i^4 \end{aligned}$$

$$+\sum_{i=1}^{N}\dot{s}_i\lambda_*\frac{3}{4}r_i^{\frac{4}{3}}(1+\phi_i(t))^{\frac{4}{3}}. \tag{29}$$

It follows from (20) that $P_dBK_d+K_d^\top B^\top P_d=-\Gamma_d$ and

$$\begin{aligned}\dot{V}_1\leq&\psi^\top[(\mathcal{SR}\Theta)\otimes(P_dA+A^\top P_d)]\psi\\&+\sum_{i=1}^{N}\frac{\lambda_*}{2}\left(s_i^2r_i^2-\frac{3}{4}r_i^2\right)\psi_i^\top(K_d^\top B^\top P_d+P_dBK_d)\psi_i\\&-\sum_{i=1}^{N}\frac{p_i}{4\lambda_*^3}\zeta_i^4\psi_i^\top(K_d^\top B^\top P_d+P_dBK_d)\psi_i.\end{aligned} \tag{30}$$

In addition, we have

$$\dot{V}_2=-\frac{\lambda_*}{8}\sum_{i=1}^{N}p_i(s_i-s_0)\psi_i^\top(K_d^\top B^\top P_d+P_dBK_d)\psi_i. \tag{31}$$

Substituting (30)–(31) into (27) yields

$$\begin{aligned}\dot{V}\leq&\psi^\top[(\mathcal{SR}\Theta)\otimes(P_dA+A^\top P_d)]\psi\\&+\sum_{i=1}^{N}\frac{\lambda_*}{2}\left(s_i^2r_i^2-\frac{1}{8}s_i-\frac{3}{4}r_i^2\right)\psi_i^\top(K_d^\top B^\top P_d\\&+P_dBK_d)\psi_i\\&+\frac{\lambda_*}{4}\sum_{i=1}^{N}p_i\left(s_0-\frac{1}{\lambda_*^4}\zeta_i^4\right)\psi_i^\top(K_d^\top B^\top P_d+P_dBK_d)\psi_i\\\leq&\psi^\top[(\mathcal{SR}\Theta)\otimes(P_dA+A^\top P_d)]\psi\\&+\sum_{i=1}^{N}\frac{1}{8}\lambda_*(s_i^2r_i^2+\tilde{s}_0)\psi_i^\top(K_d^\top B^\top P_d+P_dBK_d)\psi_i\end{aligned}$$

with $\tilde{s}_0\triangleq 2\left(s_0-\frac{\zeta_{\max}^4}{\lambda_*^4}\right)>0$. Owing to $s_0>\frac{\zeta_{\max}^4}{\lambda_*^4}+\frac{2\zeta_{\max}}{\lambda_*}$, $\frac{1}{4}\lambda_*\tilde{s}_0-\zeta_{\max}>0$, we let $\tilde{\psi}=(\sqrt{\mathcal{SR}\Theta})\psi$ and then have

$$\begin{aligned}\dot{V}\leq&\psi^\top[(\mathcal{SR}\Theta)\otimes(P_dA+A^\top P_d)]\psi\\&+\sum_{i=1}^{N}\frac{1}{4}\lambda_*s_ir_i\tilde{s}_0\psi_i^\top(K_d^\top B^\top P_d+P_dBK_d)\psi_i\\\leq&\sum_{i=1}^{N}\tilde{\psi}_i^\top[P_d(A+BK_d)+(A+BK_d)^\top P_d]\tilde{\psi}_i.\end{aligned} \tag{32}$$

By (14a), $\mathcal{X}_1^dGP_d^{-1}+(\mathcal{X}_1^dGP_d^{-1})^\top\prec 0$, that is, $P_d(A+BK_d)+(A+BK_d)^\top P_d\prec 0$. Obviously, $V=0$ if and only if $\tilde{\psi}_i=0$ for $\forall i\in\mathcal{V}$, that is, $\psi=0$. By LaSalle's invariance principle, $\psi=0$ and therefore the asymptotic consensus is achieved.

Remark 4. Similar to Remark 3, we illustrate that the SDP (14)' is feasible. Since (A,B) is stabilizable, there exists a matrix $P_d\succ 0$ such that

$$A^\top P_d+P_dA-2P_dBB^\top P_d+I=0.$$

In this case, there exist matrices K_d and $P_d\succ 0$ such that

$$(A+BK_d)^\top P_d+P_d(A+BK_d)\prec 0 \tag{33a}$$

$$P_dBK_d=(BK_d)^\top P_d\preceq 0, \tag{33b}$$

because (33a)–(33b) hold by letting $K_d=-B^\top P_d$. Along a similar line in Remark 3, it is easy to see that (14a)–(14b) hold. Noticing $\mathcal{M}Y(\mathcal{X}_0^dY)^{-1}=\mathcal{X}_1L_d$, (33b) is equivalent to

$$P_d\mathcal{M}Y(\mathcal{X}_0^dY)^{-1}=(\mathcal{M}Y(\mathcal{X}_0^dY)^{-1})^\top P_d\preceq 0, \tag{34}$$

that is, (14c') holds.

Remark 5. Compared to existing consensus protocol design methods (Deng, Zhang, & Feng, 2022; Li et al., 2015), this paper presents one of the earliest attempts to construct a scalable distributed and direct data-driven control method for addressing the asymptotic consensus problem in MASs with unknown dynamics. The challenges of unknown agent dynamics and the requirement of scalable distributed control pose significant difficulties in the co-design of the consensus protocol and adaptive schemes. To overcome these challenges, we propose the edge-based DDAC protocol for undirected communication graphs. In this protocol, the control gain and coupling matrix are directly designed using noise-free data matrices, eliminating the need for global network information and explicit knowledge of the system model. Additionally, we establish a modified node-based DDAC protocol to handle the case of directed communication graphs. Both of these proposed DDAC protocols achieve asymptotic consensus without relying on global network information or explicit knowledge of the system model. The derived results demonstrate that the proposed protocols ensure asymptotic consensus and exhibit high scalability and flexibility. The protocols are able to accommodate changes in network structure without requiring the re-computation of control and coupling gains.

Remark 6. In De Persis et al. (2024), a data collection scheme is given that is suitable for state derivatives $\dot{x}_d$. To be specific, let $\delta_d(k):=x_d\left((k+1)T_p\right)-x_d\left(kT_p\right)$, $\iota_d(k):=\int_{kT_p}^{(k+1)T_p}x_d(t)dt$, and $\varsigma_d(k):=\int_{kT_p}^{(k+1)T_p}u_d(t)dt$, and then we can obtain $\delta_d(k)=A\iota_d(k)+B\varsigma_d(k)$ where $k\geqslant 0$ and $T_p>0$ is sampling time. For a desired number of samples S, we denote $\underline{\mathcal{X}}_1^d:=[\delta_d(0)\ \delta_d(T_p)\ \dots\ \delta_d((S-1)T_p)]$, $\underline{\mathcal{X}}_0^d:=[\iota_d(0)\ \iota_d(T_p)\ \dots\ \iota_d((S-1)T_p)]$, and $\underline{\mathcal{U}}^d:=[\varsigma_d(0)\ \varsigma_d(T_p)\ \dots\ \varsigma_d((S-1)T_p)]$. Then, all the results can be restated with $\underline{\mathcal{X}}_1^d$, $\underline{\mathcal{X}}_0^d$, $\underline{\mathcal{U}}^d$ instead of $\mathcal{X}_1^d$, $\mathcal{X}_0^d$, $\mathcal{U}^d$.

4. Learning DDAC protocol with noisy data

In this section, we further consider a noisy data scenario and tackle the DDAC protocol design problem with a noisy data.

Consider again the MAS (3), but suppose that the agent d can only measure the signal z_d, that is,

$$z_d=\dot{x}_d+\omega_d \tag{35}$$

where $\omega_d:\mathbb{R}\to\mathbb{R}^n$ is an unknown measurement noise and is Lebesgue bounded in the sense that $\|\omega_d\|_\infty<+\infty$. The problem of interest is to design a DDAC protocol for the MAS (3) assuming that we measure z_d. Let

$$\begin{aligned}\mathcal{W}^d&\triangleq[\omega_d(0),\omega_d(T_p),\dots,\omega_d((S-1)T_p)]\in\mathbb{R}^{n\times S}\\\mathcal{Z}^d&\triangleq[z_d(0),z_d(T_p),\dots,z_d((S-1)T_p)]\in\mathbb{R}^{n\times S}\end{aligned} \tag{36}$$

be the unknown data matrix made of the off-line samples of ω_d, and the measured data matrix of the off-line samples of z_d. We have the next result.

Lemma 7. *Suppose that Assumption 2 holds. For any matrix $K\in\mathbb{R}^{m\times n}$, one has $A+BK=(\mathcal{Z}^d-\mathcal{W}^d)G$ where G is any solution to the system of (8).*

With noisy data, the analysis and the design of the DDAC protocol become challenging, because noisy data impact the learning phase and prevent exact identification of the dynamics. Thus, a modified DDAC protocol (11)' is constructed with (11a) and a refined adaptive scheme (12b'):

$$\dot{s}_{ij}(t)=p_{ij}h_{ij}(x_i(t)-x_j(t))^\top(x_i(t)-x_j(t)). \tag{12b'}$$

where the control gain K_d is designed in (12a). Similar to (11), we

assume the leading agent d undertakes the responsibility of computing the control gains K_d off-line, subsequently transmitting it to its neighboring agents. Employing finite-step transmission, all agents can successfully receive K_d. The above process will be carried out in the initial stage.

We introduce the set

$$\mathcal{W}_0 \triangleq \{\mathcal{W} : \mathcal{W}\mathcal{W}^\top \prec \Lambda\Lambda^\top\} \tag{38}$$

where Λ is a known matrix, and we make the next assumption.

Assumption 5. $\mathcal{W}^d \in \mathcal{W}_0$.

The selection of the set $\mathcal{W}_0$ is indicative of our a priori knowledge or estimation regarding the noise ω_d. For example, if it is known that there exists a positive constant $d_* > 0$ such that $\|\omega_d\|_\infty \leq d_*$, as demonstrated in De Persis et al. (2024), Qi, Liu, Wang, and Sun (2022), then we may set $\Lambda \triangleq d_*\sqrt{SI}$, where S denotes the number of samples.

Theorem 8. *Consider a general linear MAS (3). Suppose that Assumptions 2–3 and 5 hold with a given Λ. There exist $\varepsilon > 0$, $\epsilon > 0$ and the decision variable $Y \in \mathbb{R}^{S\times n}$ satisfying (14)" with the form of (15a"), (14b) and (15c")*

$$\begin{bmatrix} \mathcal{Z}^d Y + (\mathcal{Z}^d Y)^\top + \varepsilon\Lambda\Lambda^\top & Y^\top \\ Y & -\varepsilon I \end{bmatrix} \prec 0 \tag{15a"}$$

$$\begin{bmatrix} \mathcal{M}_1 Y + (\mathcal{M}_1 Y)^\top + \epsilon\Lambda\Lambda^\top & Y^\top \\ Y & -\epsilon m_d^{-1} I \end{bmatrix} \prec 0 \tag{15c"}$$

where $\mathcal{M}_1 = \mathcal{Z}^d M_*$ with $M_* = M_d M_v \mathbb{I}_d \mathcal{U}^d$, and m_d is the largest eigenvalue of $M_*^\top M_*$. For any initial condition, the DDAC protocol (11)' guarantees that the MAS (3) with general linear dynamics achieves asymptotic consensus.

Proof. Consider a Lyapunov function candidate as

$$V = \sum_{i=1}^N \xi_i^\top P_d \xi_i + \sum_{i=1}^N \sum_{j=1}^N h_{ij} \frac{(m_1 s_{ij} - m_2 s_0)^2}{4 m_1 p_{ij}}$$

with $P_d = (\mathcal{X}_0^d Y)^{-1}$, s_{ij} and p_{ij} given in (11a)–(11b) and a positive number s_0 satisfying $s_0\lambda_2 > 1$. m_1 and m_2 are the smallest and largest eigenvalues of $-(P_d BK_d + K_d^\top B^\top P_d)$, respectively. It follows from (3) and (35)–(36) that

$$\mathcal{Z}^d - \mathcal{W}^d = A\mathcal{X}_0^d + B\mathcal{U}^d = \begin{bmatrix} B & A \end{bmatrix} M_d^\top$$

with M_d in (10). By (10) and (12b), one has

$$(\mathcal{Z}^d - \mathcal{W}^d) L_d = \begin{bmatrix} B & A \end{bmatrix} M_d^\top M_d M_v \mathbb{I}_d K_d = BK_d. \tag{39}$$

Note that (15c") is equivalent to

$$\mathcal{M}_1 Y + (\mathcal{M}_1 Y)^\top + \epsilon\Lambda\Lambda^\top + \epsilon^{-1} Y^\top M_*^\top M_* Y \prec 0.$$

It follows from De Persis et al. (2024, Lemma 6) that for $\forall\ \mathcal{W}^d \in \mathcal{W}_0$,

$$\mathcal{M}_1 Y + (\mathcal{M}_1 Y)^\top - \mathcal{W}^d M_* Y - (\mathcal{W}^d M_* Y)^\top \prec 0,$$

that is, $(\mathcal{X}_0^d Y)^{-1}(\mathcal{Z} - \mathcal{W}^d) M_* Y (\mathcal{X}_0^d Y)^{-1} + ((\mathcal{X}_0^d Y)^{-1}(\mathcal{Z} - \mathcal{W}^d) M_* Y (\mathcal{X}_0^d Y)^{-1})^\top \prec 0$, equivalently,

$$(\mathcal{X}_0^d Y)^{-1}(\mathcal{Z} - \mathcal{W}^d) L_d + [(\mathcal{X}_0^d Y)^{-1}(\mathcal{Z} - \mathcal{W}^d) L_d]^\top \prec 0.$$

It follows from (39) that $P_d BK_d + K_d^\top B^\top P_d \prec 0$.

The derivative of V along the trajectory of the system (3) can be obtained as

$$\begin{aligned}
\dot V &= \sum_{i=1}^N [\xi_i^\top (P_d A + A^\top P_d)\xi_i \\
&\quad + \xi_i^\top (P_d BK_d + K_d^\top B^\top P_d) \sum_{j=1}^N s_{ij} h_{ij} \psi_{ij}] \\
&\quad + \sum_{i=1}^N \sum_{j=1}^N h_{ij} \frac{(m_1 s_{ij} - m_2 s_0)}{2} \psi_{ij}^\top \psi_{ij} \\
&\leq \sum_{i=1}^N [\xi_i^\top (P_d A + A^\top P_d)\xi_i \\
&\quad + \xi_i^\top (P_d BK_d + K_d^\top B^\top P_d) \sum_{j=1}^N s_{ij} h_{ij} \psi_{ij}] \\
&\quad - \sum_{i=1}^N \sum_{j=1}^N \frac{h_{ij} s_{ij}}{2} \psi_{ij}^\top (P_d BK_d + K_d^\top B^\top P_d)\psi_{ij} \\
&\quad + \sum_{i=1}^N \sum_{j=1}^N \frac{h_{ij} s_0}{2} \psi_{ij}^\top (P_d BK_d + K_d^\top B^\top P_d)\psi_{ij}
\end{aligned} \tag{40}$$

By substituting (18) into (40), we obtain

$$\begin{aligned}
\dot V &= \sum_{i=1}^N \xi_i^\top (A^\top P_d + P_d A)\xi_i \\
&\quad + s_0 \xi^\top [\mathcal{L} \otimes (P_d BK_d + K_d^\top B^\top P_d)]\xi \\
&\leq \sum_{i=1}^N \xi_i^\top [A^\top P_d + P_d A + s_0\lambda_2 (P_d BK_d + K_d^\top B^\top P_d)]\xi_i \\
&\leq \sum_{i=1}^N \xi_i^\top [(A + BK_d)^\top P_d + P_d (A + BK_d)]\xi_i.
\end{aligned}$$

where the second inequality holds due to $s_0\lambda_2 > 1$.

Let $G = YP_d$, and then $K_d = \mathcal{U}^d G$ and $\mathcal{X}_0^d G = I$, which satisfies (8), and thus $A + BK_d = (\mathcal{Z}^d - \mathcal{W}^d)G$ according to Lemma 7. By (15a"), one has

$$\begin{aligned}
\mathcal{Z}^d G P_d^{-1} + (\mathcal{Z}^d G P_d^{-1})^\top + \varepsilon\Lambda\Lambda^\top \\
+ \varepsilon^{-1} (G P_d^{-1})^\top G P_d^{-1} \prec 0,
\end{aligned}$$

and it follows from De Persis et al. (2024, Lemma 6) that for $\forall\ \mathcal{W}^d \in \mathcal{W}_0$,

$$\mathcal{Z}^d G P_d^{-1} + (\mathcal{Z}^d G P_d^{-1})^\top - \mathcal{W}^d G P_d^{-1} - (\mathcal{W}^d G P_d^{-1})^\top \prec 0.$$

This implies that $P_d(\mathcal{Z}^d - \mathcal{W}^d)G + ((\mathcal{Z}^d - \mathcal{W}^d)G)^\top P_d \prec 0$, that is, $(A + BK_d)^\top P_d + P_d(A + BK_d) \prec 0$. Thus, there exists a positive number μ_1 such that $\dot V \leq -\mu_1 \sum_{i=1}^N \xi_i^\top \xi_i$. This implies $\dot V \leq 0$ and thus V is monotonously non-decreasing. Note that $\dot V = 0$ if and only if $\xi_i = 0$ for $i = 1, \ldots, N$. This implies that $\lim_{t\to+\infty} \xi_i(t) = 0$, and the asymptotic consensus can be achieved eventually.

Remark 7. Most existing model-free methods (Bu et al., 2017; Jiang & He, 2019; Wang, Jin, Mao, Vasilakos, & Tang, 2020; Zhang et al., 2019) follow a two-step control procedure involving system identification and control design. However, system identification can be challenging and complex, especially when there is insufficient information available and the noise is involved in the collected data. In contrast, the proposed DDAC protocol (11) takes a different approach via bypassing the system identification process and directly utilizing data matrices $\mathcal{U}^d$, $\mathcal{X}_0^d$ and $\mathcal{Z}^d$ (or $\mathcal{X}_1^d$) to design the control gain K_d or the coupling matrix Γ_d. This direct data-driven control method does not require the identification of system matrices A and B, thereby reducing the computational complexity associated with system identification.

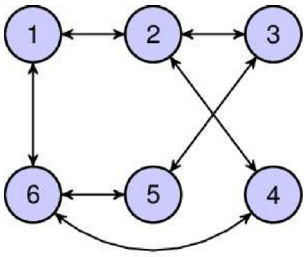


Fig. 1. Communication graph $\mathcal{G}$ with 6 nodes where each node represents an agent.

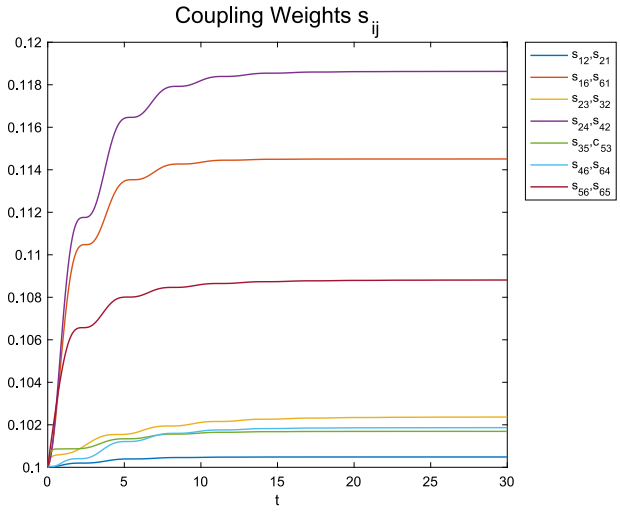


Fig. 2. Coupling weights of each edge under the adaptive scheme (11b).

5. Simulation

In this section, three examples are provided to verify the effectiveness of our DDAC protocols.

Example 2. Consider an MAS with six agents that are connected through a undirected graph (see Fig. 1). The dynamics of each agent is described as (3) with

$$A=\begin{bmatrix}0 & -1 & 0\\ 1 & 0 & 0\\ 0 & 0 & -3\end{bmatrix},\quad B=\begin{bmatrix}1\\0\\0\end{bmatrix}. \tag{41}$$

Suppose that the matrices A and B are unknown. We consider distributed DDAC protocol (11) for system (3). To this end, we collect the data by running an experiment with the data-generating input and initial state that are sampled uniformly from $u_d, x_d \in [0, 1]$. We collect 10 samples which is sufficient for Assumption 1 to hold, that is, $\{u_d(s\cdot T_p), x_d(s\cdot T_p), \dot{x}_d(s\cdot T_p)\}_{s=0}^{9}$ with the interval $T_p = 0.1$. By solving SDP (14), one obtains

$$K_d=\begin{bmatrix}-1.7002 & -0.3034 & 1.9522\end{bmatrix} \tag{42}$$

$$\Gamma_d=\begin{bmatrix}0.0359 & 0.0064 & -0.0412\\ 0.0064 & 0.0011 & -0.0074\\ -0.0412 & -0.0074 & 0.0473\end{bmatrix} \tag{43}$$

For $\forall(i,j)\in\mathcal{E}$, let $s_{ij}(0)=s_{ji}(0)=0.1$, and the evolutions of s_{ij} are shown in Fig. 2. It can be seen from Figs. 3 and 4 that the proposed DDAC protocol (11) is able to ensure that the trajectories of all agents converge to the same.

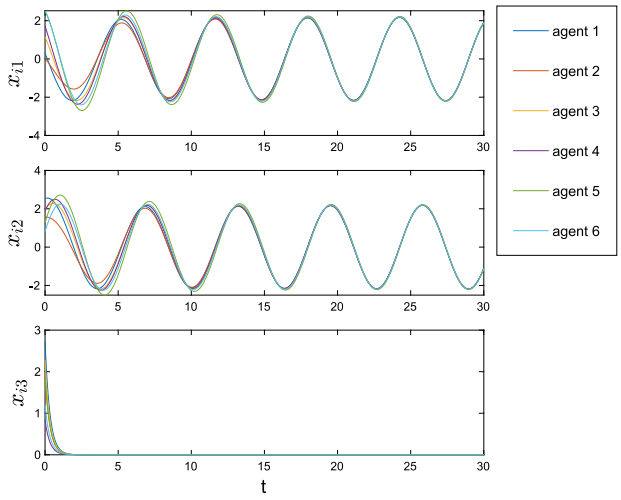


Fig. 3. Trajectories of components x_{i1}, x_{i2} and x_{i3} of six agents under the DDAC protocol (11).

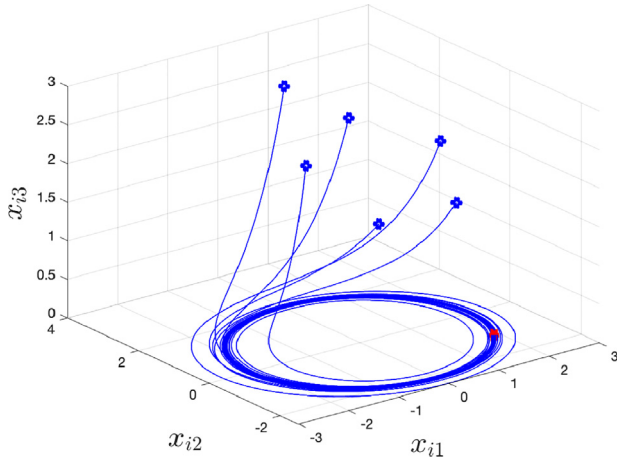


Fig. 4. State trajectories of six agents under the DDAC protocol (11).

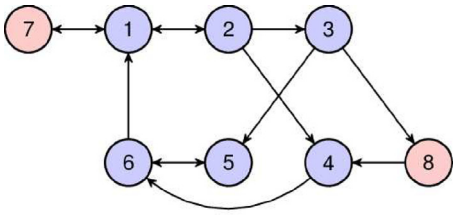


Fig. 5. Communication graph $\mathcal{G}$, where the original graph includes 6 nodes and nodes 7 and 8 are newly added.

Example 3. Consider an MAS with a directed graph (see Fig. 5), and the dynamics of each agent is described as (3) with

$$A=\begin{bmatrix}-1 & -3\\ 0 & 0\end{bmatrix},\quad B=\begin{bmatrix}0\\2\end{bmatrix}. \tag{44}$$

Suppose that the matrices A and B are unknown. We discuss the distributed DDAC protocol (25) for system (3). To this end, we collect the data by running an experiment with the data-generating input and initial state that are sampled uniformly from $u_d, x_d \in [0, 1]$. We collect 10 samples which is sufficient for Assumption 2 to hold, that is, $\{u_d(s\cdot T_p), x_d(s\cdot T_p), \dot{x}_d(s\cdot T_p)\}_{s=0}^{9}$ with the interval $T_p = 0.1$. By solving SDP (14)', one obtains

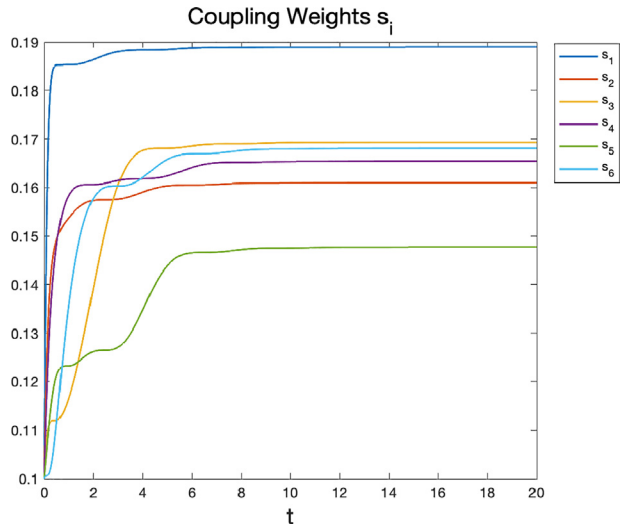


Fig. 6. Coupling weights of each node under the adaptive scheme (25b).

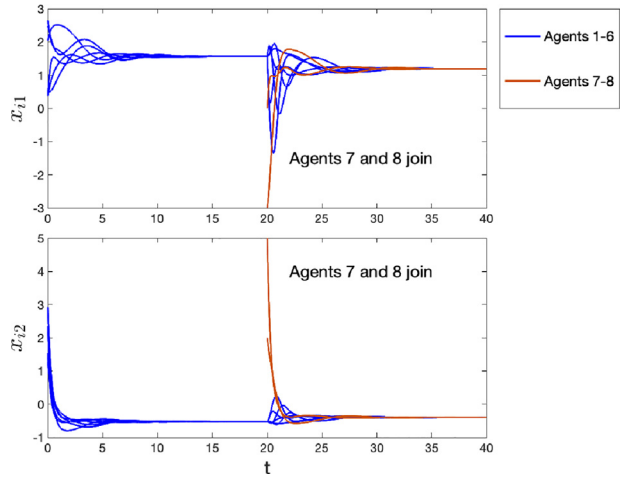


Fig. 7. Trajectories of components x_{i1} and x_{i2} of eight agents under the DDAC protocol (25).

$$K_d = \begin{bmatrix} 2.3148 & -3.2111 \end{bmatrix},$$
$$\Gamma_d = \begin{bmatrix} 0.0177 & -0.0246 \\ -0.0246 & 0.0341 \end{bmatrix}, P_d = \begin{bmatrix} 0.0177 & -0.0246 \\ -0.0246 & 0.0341 \end{bmatrix}.$$

For $\forall i \in \mathcal{V}$, let $s_i(0) = 0.1$, and the evolutions of s_i are shown in Fig. 6. It can be seen from Figs. 7 and 8 that the proposed DDAC protocol (25) is able to ensure that the trajectories of all agents converge to the same. In addition, when new agents 7 and 8 join in, the DDAC protocol can be self-adjusted with local information without redesigning the protocol and its parameters. Then, a new consensus state is generated and all agents converge to the new one.

Example 4. In this example, we consider the case of noisy data. Consider an MAS with $N = 5$ agents that are connected through a circle graph. The dynamics of each agent is described as (3) with

$$A = \begin{bmatrix} 0 & 0.7 \\ -0.7 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}. \tag{45}$$

Suppose that the matrices A and B are unknown. We consider distributed DDAC protocol (11)' for system (3). To this end, we

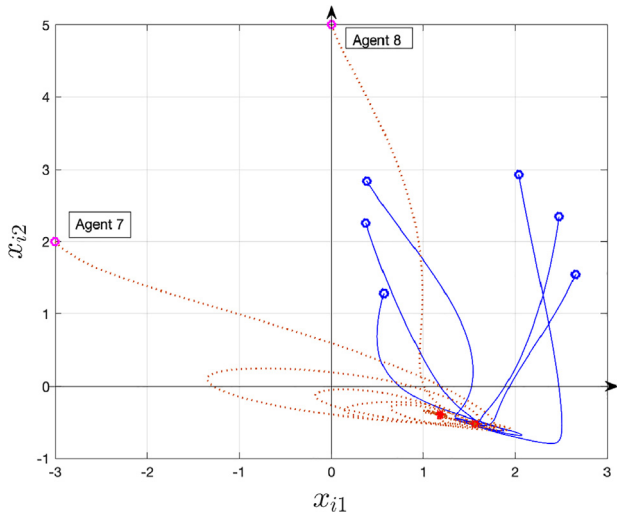


Fig. 8. State trajectories of eight agents under the DDAC protocol (25).

collect the noisy data by running an experiment with the data-generating input and initial state that are sampled uniformly from $u_d, x_d \in [0, 10]$. We collect 50 samples which is sufficient for Assumption 2 to hold. The noise is taken as a random sequence within $[-0.1, 0.1]$. We use $\Lambda = 0.1\sqrt{SI}$. To solve ((14)''), we used CVX and obtain

$$K_d = \begin{bmatrix} 5.8836 & -8.7250 \\ 5.9551 & 9.3526 \end{bmatrix} \tag{46}$$

For $\forall (i, j) \in \mathcal{E}$, let $s_{ij}(0) = s_{ji}(0) = 0.1$. It can be seen from Fig. 9(a) that the proposed DDAC protocol (11)' is able to ensure that the trajectories of all agents converge to the same.

Then, in order to discuss the scalability of the DDAC protocol (11)' that learning from noisy data, we further consider the following two cases: (i) $N = 10$ agents where five new agents join the system [see Fig. 9(b)]; and (ii) $N = 20$ vehicles with fifteen new vehicles joining the system [see Fig. 9(c)]. Each agent adjusts its trajectory based on (11)'. It is evident from Fig. 9 that, despite the increasing number of participating agents (from $N = 5$, to $N = 10$, and finally to $N = 20$), all agents' trajectories asymptotically converge to the same, obviating the necessity for reconfiguring any gains or parameters. The coupling weights s_{ij} exhibit the capability for online self-adjustment with arbitrary initial values (see Fig. 10).

6. Conclusion

In this paper, we have presented a scalable distributed Data-Driven Adaptive Consensus (DDAC) protocol to address the consensus problem in Multi-Agent Systems (MASs) with general linear unknown dynamics. We have considered both undirected and directed communication graphs in our analysis. The proposed protocol utilizes a direct data-driven approach to design the control gain and coupling matrix, enabling the development of edge/node-based adaptive schemes and DDAC protocols using available noiseless and noisy data. We have demonstrated that the proposed protocols are capable of ensuring asymptotic consensus in MASs with unknown dynamics while exhibiting high scalability. Through theoretical analysis and proofs, we have established the effectiveness and feasibility of the protocols. Furthermore, we have provided simulation examples to validate the performance and efficacy of our proposed protocols. In summary, our proposed data-driven approach and protocols offer practical

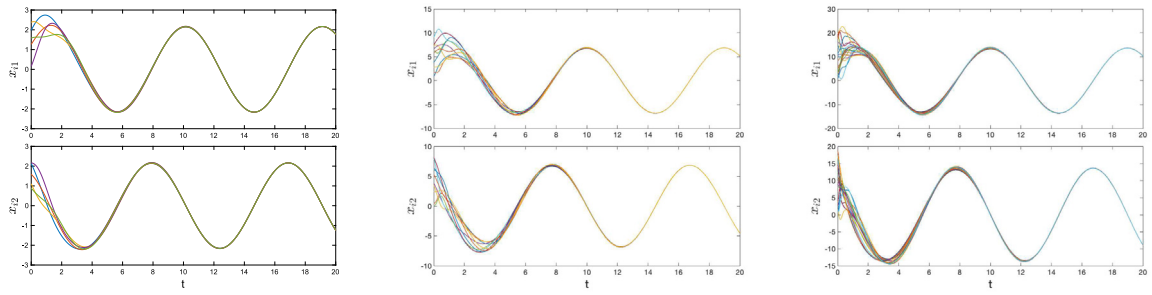


Fig. 9. Trajectories of components x_{i1} and x_{i2} of agents under the DDAC protocol (11)' where three cases of $N = 5$ (a), $N = 10$ (b) and $N = 20$ (c) are displayed, respectively.

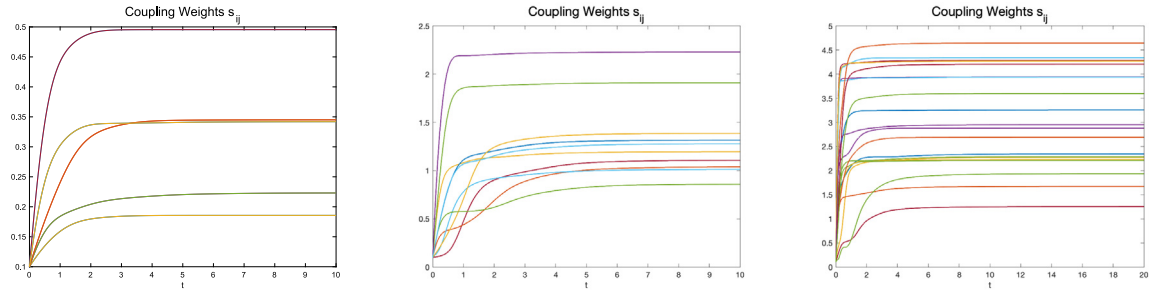


Fig. 10. Coupling weights of each edge determined by the adaptive scheme (12b'), where three cases of $N = 5$ (a), $N = 10$ (b) and $N = 20$ (c) are displayed, respectively.

solutions that do not require explicit system model identification, showcasing their effectiveness and scalability. One of the future research topics is to extend the main results of this paper to more complicated systems such as nonlinear MASs and stochastic systems (Caballero-Águila, Hermoso-Carazo, & Linares-Pérez, 2017, 2020; Ciuonzo, Aubry, & Carotenuto, 2017; Ding et al., 2018; Shmaliy, Lehmann, Zhao, & Ahn, 2018) and general networks such as switching topologies (Meng & Jia, 2015; Xu, Cao, Yu, & Lu, 2014).

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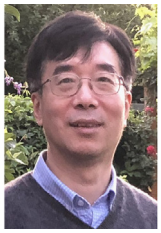
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